Singapore School Kinderland

Secondary 2 · Computer Science · Term Plan · Term 2

	Ch 1.3: Data Storage & Compression (Pg 32-44)				Ch 2.2 & 2.3: Error Detection & Encryption (Pg 54-74)				Ch 1.3 Review & Practice		Ch 2 Review & Practice		Term Review		Exam
	Wk 1		Wk 2		Wk 3		Wk 4		Wk 5		Wk 6		Wk 7		Wk 8
	Lesson Objective	Content	Lesson Objective	Content	Lesson Objective	Content	Lesson Objective	Content	Lesson Objective	Content	Lesson Objective	Content	Lesson Objective	Content	
Day 1	St will define storage units: bits, bytes, KB/MB/GB (powers of 10) vs KiB/MiB/GiB (powers of 2), and convert between storage units.	TB Pg 32-34 1.3.1 Storage Units KiB vs KB	St will calculate sound file sizes (Sample Rate x Resolution x Duration x Channels) and solve practical calculations.	TB Pg 38-40 1.3.3 Sound File Size Math WS Sound Tasks	St will analyze Even and Odd Parity checks and solve Parity Block grids to locate corrupted bit positions during transmission.	TB Pg 54-57 2.2.1-2 Parity Blocks & Correction WS Matrix	St will explain Check Digits and calculate check digits using Modulo 11 and ISBN-13 algorithms to detect entry errors.	TB Pg 60-61 2.2.5 Check Digits & Modulo 11 WS Math Tasks	St will compare Symmetric vs Asymmetric encryption, explaining key exchange challenges and brute force vulnerabilities.	TB Pg 63-67 2.3.1-2 Symmetric Encryption & Keys	St will solve Cambridge IGCSE past paper math problems on file sizes (image/sound) and RLE text/image compression.	WS Pg 1-4 Cambridge Past Paper File Size & RLE Tasks	St will complete comprehensive review of storage calculations, compression trade-offs, parity block matrices, and encryption pairs.	WS Comprehensive Theory & Math Review Group Q&A	Term Exam
Day 2	St will calculate image file sizes (Resolution x Colour Depth) in bytes, KiB, and MiB with structured formula practice.	TB Pg 35-37 1.3.2 Image File Size Math WS Image Tasks	St will compare Lossy vs Lossless compression, identify suitable file types (JPEG/MP3 vs PNG/FLAC), and apply Run-Length Encoding (RLE).	TB Pg 41-44 1.3.4 Lossy/Lossless & RLE WS Compression	St will explain Checksum algorithms (block sum calculation at source and destination) and evaluate Echo Check limitations.	TB Pg 58-60 2.2.3-4 Checksum & Echo Checks WS Case Study	St will evaluate Automatic Repeat Requests (ARQ): sequence of ACK, NAK, timeouts, and automatic re-transmissions.	TB Pg 61-62 2.2.6 ARQ Protocol Mechanics WS Flowcharts	St will explain Public/Private key pairs, Hashing, Digital Signatures, and SSL/TLS Digital Certificates for secure e-commerce.	TB Pg 68-74 2.3.3-4 Asymmetric Encryption & Signatures	St will solve Cambridge past paper questions on error detection protocols and encryption key security scenarios.	WS Pg 5-8 Cambridge Past Paper Error & Encryption Tasks	St will take a timed mock test covering all Paper 1 question styles from Ch 1.3, Ch 2.2, and Ch 2.3.	WS Term 2 Mock Exam Assessment	Results & Feedback